PRODUCT REQUIREMENT DOCUMENT

PRD: Field Service Management Platform

PRD-2026 · AIAvenue Agency

Executive Summary

The Field Service Management (FSM) Platform is a cloud-native, real-time field operations management system built on AWS Amplify Gen 2 with a React + Vite + TailwindCSS frontend. It enables businesses to dispatch, track, and manage field technicians through an intuitive Kanban-based task board with role-based access control (Admin/Technician). The platform ingests orders via a Shopify webhook pipeline (Shopify → Lambda → DynamoDB → Frontend), synchronizes state across all users via 30-second polling against AppSync + DynamoDB, and provides an executive dashboard powered by Recharts. The system is deployed through a fully automated CI/CD pipeline on GitHub with a single-page application (SPA) 404 redirect fix for AWS Amplify Hosting. Target customers are small-to-medium enterprises (SMEs) in India managing 10–200 field service technicians across retail, appliance repair, and last-mile delivery verticals.

Problem Statement

Core Problem

Field service businesses in India rely on fragmented tools — spreadsheets for scheduling, WhatsApp for dispatching, and manual phone calls for status updates. No single affordable solution unifies order ingestion, technician assignment, real-time status tracking, and performance analytics.

User Pain Points

- **Dispatch Chaos:** Dispatchers spend 2–3 hours daily manually assigning orders to technicians via phone or WhatsApp groups, leading to assignment errors and delayed response times.
- **No Real-Time Visibility:** Business owners have zero visibility into whether a technician has started, completed, or delayed a job. Customer status updates rely on the technician remembering to call back.
- **Order Sourcing Fragmentation:** Orders arrive from Shopify, phone calls, walk-ins, and email. No unified pipeline exists to ingest and normalize orders into a single queue.
- **Role Confusion:** Without role-based access, technicians see all orders and can accidentally modify assignments meant for other teams.
- **Delayed Billing & Reporting:** At the end of each month, managers manually reconcile completed jobs against invoices. Missed billings routinely reach 5–8% of revenue.
- **Scalability Ceiling:** Spreadsheets break beyond 20 concurrent jobs. Businesses hit a hard ceiling on how many technicians they can manage.

User Segments Affected

- Operations managers / dispatchers
- Field technicians
- Business owners / stakeholders
- Customers (indirect — receive better ETAs and status updates)

User Research & Personas

Persona 1: Rajesh Mehta — Operations Manager

Attribute	Detail
Age	42
Role	Operations Manager at a home appliances repair company
Tech Comfort	Moderate — comfortable with web apps, not with APIs or scripts
Team Size	Manages 35 technicians across 3 cities
Daily Workflow	Starts at 7 AM taking WhatsApp calls; spends 3 hours dispatching; calls each technician at 2 PM for status; reconciles job sheets at 6 PM
Pain Points	Cannot tell which technician is free; double-assigns jobs by accident; loses paperwork
Needs	Single dashboard showing all jobs, real-time technician status, automatic order import from Shopify

Persona 1: Rajesh Mehta — Operations Manager

Persona 2: Sunil Kumar — Field Technician

Persona 3: Priya Sharma — Business Owner

Market Analysis

Total Addressable Market (TAM)

The global field service management market was valued at **$5.1 billion in 2024** and is projected to reach **$8.7 billion by 2029** (CAGR 11.2%). India's share, at approximately **6–7% of global SaaS spend**, yields a TAM of **₹2,550–3,000 crore**.

Serviceable Addressable Market (SAM)

SMEs in India with 10–200 field technicians across retail after-sales, appliance repair, electronics maintenance, and last-mile delivery constitute ~34,000 businesses. At an average willingness-to-pay of ₹5,000–8,000/month, the SAM is approximately **₹245–272 crore/year**.

Serviceable Obtainable Market (SOM)

With a focused go-to-market in Tier-1 and Tier-2 cities and a product-led growth strategy targeting 500 subscriptions in Year 1 ramping to 3,000 by Year 3, the SOM is **₹3.6 crore in Year 1** and **₹21.6 crore in Year 3**.

Market Rationale

- India's service sector contributes 55% to GDP with rapidly digitising SME base
- Government push for digital payments and GST compliance indirectly drives software adoption
- Post-COVID labour shortage makes workforce optimisation tools essential
- Shopify's merchant base in India grew 3× between 2020–2024 — direct pipeline opportunity

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Competitive Analysis

Competitor	Strengths	Weaknesses	Our Edge
Zoho Field Service	Mature product, broad CRM integration, AI scheduling	High price (₹15,000+/month), complex setup, overkill for SMEs	Lower price point (₹5,000–8,000), simpler UX, Shopify native integration
HouseCall Pro	Excellent mobile app, customer communication tools	US-focused (no India-specific pricing or support), no Shopify webhook	Indian market focus, local support, INR pricing, GST invoices
Servicemax (Salesforce)	Enterprise-grade, IoT integration, asset management	Extremely expensive (₹50,000+/month), requires Salesforce license	5–10× cheaper, self-serve onboarding, no vendor lock-in
FieldEZ	India-native, strong in manufacturing	Outdated UI, limited API ecosystem, poor mobile experience	Modern React stack, clean UX, real-time Kanban, Shopify first-party integration
Manual / Spreadsheets	Zero cost, high flexibility	No real-time sync, error-prone, no analytics, no role-based access	Automation, accuracy, real-time visibility, measurable ROI

Solution Overview

Key Features

1. **Unified Order Inbox:** All orders ingested via Shopify webhook (Shopify → Lambda → DynamoDB), displayed in a single queue with order details, customer info, and line items 2. **Kanban Task Board:** Four-column workflow (Pending → Accepted → Start → Complete) with drag-and-drop assignment. Technicians see only their assigned tasks 3. **Role-Based Access Control (RBAC):** - **Admin:** Full CRUD on orders, technician management, dashboard access, user creation - **Technician:** View assigned tasks, change status (Accept → Start → Complete), read-only product info 4. **30-Second Polling Sync:** Clients poll AppSync GraphQL endpoint every 30 seconds for state changes. No WebSocket overhead 5. **Executive Dashboard:** Recharts-based visualisations — daily job volume, average completion time, technician utilisation, revenue tracking, bottleneck alerts 6. **CI/CD Pipeline:** GitHub → Amplify Hosting. SPA 404 fix via redirect rules in `amplify.yml` 7. **Auth:** Cognito User Pools with hosted UI. Email + password authentication. Admin group vs. Technician group

Architecture

```
Shopify ──webhook──> API Gateway ──> Lambda (transform + store) ──> DynamoDB
                                                                      │
                                                                  AppSync (GraphQL)
                                                                  ┌──────┴──────┐
                                                                  30s poll  30s poll
                                                                  Admin UI  Technician UI
                                                                  (React)   (React)
                                                                  │         │
                                                                  Cognito Auth  Cognito Auth
```

Tech Stack

Layer	Technology
Frontend	React 18 + Vite + TailwindCSS + shadcn/ui
State Management	Zustand
Auth	Amazon Cognito User Pools
API Layer	AWS AppSync (GraphQL)
Database	Amazon DynamoDB

Webhook Pipeline	AWS Lambda + API Gateway
Hosting	AWS Amplify Hosting
CI/CD	GitHub Actions
Charts	Recharts
Icons	Lucide React

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Shopify ──webhook──> API Gateway ──> Lambda (transform + store) ──> DynamoDB │ AppSync (GraphQL) ┌──────┴──────┐ 30s poll 30s poll Admin UI Technician UI (React) (React) │ │ Cognito Auth Cognito Auth
```

Tech Stack

Feature Deep-Dive (RICE Priority Matrix)

Feature	Reach (users/month)	Impact (1–5)	Confidence (%)	Effort (person-weeks)	RICE Score
Shopify webhook order ingestion	500	5	95%	3	791.7
Kanban board with drag-and-drop	500	5	90%	4	562.5
Role-based access (Admin / Technician)	500	4	95%	2	950.0
30-second polling sync	500	5	90%	2	1125.0
Recharts executive dashboard	50 (admins only)	4	80%	3	53.3
Order detail modal with customer info	500	3	90%	1.5	900.0
Technician mobile responsive view	400	5	85%	3	566.7
SPA 404 redirect fix	500 (all users)	5	99%	0.5	4950.0
CI/CD pipeline setup	— (dev team)	5	95%	1	—
Cognito auth integration	500	5	95%	2	1187.5

Formula: RICE = (Reach × Impact × Confidence) / Effort

Top Priority Features (by RICE Score): 1. SPA 404 redirect fix (4950) 2. Cognito auth integration (1187.5) 3. 30-second polling sync (1125.0) 4. Order detail modal (900.0) 5. Role-based access (950.0)

Success Metrics (KPIs)

KPI	Target (Month 6)	Measurement Method
Order processing time	< 2 minutes from Shopify webhook → Kanban board	CloudWatch log analysis
Dashboard adoption rate	> 80% of admin users logging in daily	Cognito + Amplify Analytics
Technician status update rate	> 90% of jobs have start + complete timestamps	DynamoDB query
Sync latency (p95)	< 35 seconds end-to-end	AppSync metrics
Deployment success rate	> 99% CI/CD pipeline green builds	GitHub Actions
NPS (Admin users)	> 45	Quarterly survey
Billing leakage recovery	> ₹50,000/month per 100-technician company	Admin-reported reconciliation

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Roadmap

Phase 1 — Foundation (Weeks 1–4)

- AWS Amplify Gen 2 scaffolding + Cognito auth
- DynamoDB schema design + AppSync GraphQL setup
- React + Vite + TailwindCSS skeleton with shadcn/ui
- Zustand store with auth slice
- SPA 404 redirect fix in `amplify.yml`
- CI/CD pipeline with GitHub Actions

Phase 2 — Core Features (Weeks 5–10)

- Shopify webhook pipeline (Lambda + API Gateway)
- Order ingestion and DynamoDB write
- Kanban board with 4 columns (Pending → Accepted → Start → Complete)
- Drag-and-drop status transitions
- Role-based access (Admin / Technician views)
- 30-second AppSync polling sync
- Technician mobile-responsive layout

Phase 3 — Analytics & Launch (Weeks 11–14)

- Recharts executive dashboard (6 widgets)
- Order detail modal with full customer info

- Technician performance metrics tab
- Load testing (100 concurrent users)
- UAT with 3 beta customers
- Production launch

Phase 4 — Post-Launch (Month 4–6)

- In-app notifications for status changes
- Customer ETA sharing (SMS/WhatsApp)
- Weekly email digest for admins
- Offline PWA support for technicians
- Billing reconciliation export (CSV/PDF)

Impact Estimate

Quantitative Impact (Annualised, per 100-technician company)

Metric	Before	After	Improvement
Dispatch time (daily)	3 hours	20 minutes	89% reduction
Missed billings	₹6,00,000/year	₹60,000/year	90% reduction
Customer wait time (phone)	15 min avg.	30 sec (in-app)	97% reduction
Jobs completed/day	18/technician	24/technician	33% increase
Manager time on status checks	2.5 hours/day	10 min/day	93% reduction

Quantitative Impact (Annualised, per 100-technician company)

ROI for Customer (₹/year)

ROI for Platform (₹/year)